seat, which is the critical point referred to by the committee, was 6⅜ in. Several years later the wheel seat for this axle was increased to 6½ in.

In paragraph 15, the committee brings out the point that the 6½-in. wheel seat is the standard diameter of the 5 in. by 9 in. axle. That is true, but the rules of interchange and the table in Exhibit A, allow the wheel seat of a 5 in. by 9 in. axle to be made as low as 6¼ in.

In paragraph one, on page one, they speak of the failures being due to overloading. The question of failure from overloading is largely a question of experience, in so far that all of our stresses in shapes or structures, have been determined upon someone's experience. In all the twenty years' service of these axles, there has never been a single report of an axle breaking in the wheel seat, even though a large number of them were originally 6⅜ in.

I wish to ask that this additional non-standard M. C. B. axle be put in this list, in accordance with the letter to the committee, with one modification. We stated in the letter that we desire 152,000 lb. as the load. We would like to modify that and use 148,000 lb., as that will take care of the real carrying capacity of the cars—110,000 lb. in most cases—and allow, at the same time, throwing the axle stresses well within the limits as established in the standard M. C. B. axles.

I wish to illustrate that by referring to the variations of weight and comparing this axle with the standard M. C. B. 5-in. by 9-in. axle and the standard M. C. B. 5½-in. by 10-in. axle. I will make the first comparison with the 5½-in. by 10-in. axle. The decrease in weight is 8 per cent of the decrease in the strength of the axle at the wheel seat is 10.7 per cent. That looks as if it was below the standard axle, but you have further to take into consideration that the distance from the center of the bar to the center of the rail, measured in a horizontal line, is reduced one-half inch, which is equivalent to a 5.5 per cent reduction; so if we add the 5½ per cent to the 8 per cent, we have practically 13 per cent reduction in stresses and 10.7 per cent in reduction of the strength of the axle at that point. It is, therefore, well within the limit at the wheel seat.

Comparing it again with the 5¼-in. by 9-in. axle, using the 148,000 lb. and 132,000 lb., the lengths there are identical. The increase in lading is 12.1 per cent. The increase in the strength of the axle at the wheel seat is 12.5 per cent. As a matter of safety and strength, it meets all the requirements of the M. C. B. axle.

I therefore move that this non-standard M C.. B. axle be placed in the iist with the others.

As to the operation of the rule, it is significant to the company that they have now in service, so far as we know, something over 14,000 open-top coal cars, so constructed. None of these cars have been built since 1909, so that you will see we started to be good 10 years ago. By cutting down the capacity of these cars on an average of 10 tons, that would be equivalent to a destruction of 2,800 50,000-lb. cars, the value of which, if you put it on a liberal basis, would be something like $7,000,000. If we should estimate the carrying capacity of the cars on the basis of mileage of the cars during the year 1916, of which we have a record, and allowing only half the mileage to be the loaded mileage, it would reduce the capacity something over 1,000,000,000 ton-miles.

R. L. Kleine (Penna. Lines): The present chaotic condition of the markings of cars and the amount of loaded carrying capacity has brought about a recommendation on the part of the U. S. R. R. A. Car Service Section to get a uniform marking, in order to obtain the maximum loaded carrying capacity on the cars, and as well to overcome the trouble that has been experienced with broken axles in service. The matter was referred to the Car Truck Committee to solve the proposition.

We well know that the M. C. B. standard axle has been abused in that it has been turned down below the safe limits set for the particular axle, and I do not think we can justify Mr. Hennessey's statement that, due to expediency, we have disregarded the safe loading on the axle and turned it down below the safe limits, and ask to continue such practice.

The committee has taken all of these details into consideration, and formulated the safe loads on the standard axles, as well as figuring the non-standard axles by the same formula as the standard axle was figured, and in cars now in service, we do not think that the standard axle will have to be marked down to the figures presented by the committee.

The committee has had a joint conference with the Car Service Section, and also with the Traffic Section, and they have unanimously agreed to the markings as recommended by the committee.

In regard to submitting to the car construction committee such questions as marking down the cars below the carrying capacity, the thought, there, was that the controversies might arise where a car has been marked in accordance with the standard axle under the car, and the car shows weakness in service. When such occasion arises there will be a controversy between the road on which the car failed and the car owner, and it was thought best that such controversies should be submitted to the car construction committee.

In regard to the Norfolk & Western axle, the committee carefully considered the subject. This is a special axle of the same length as the 5-in. by 9-in. axle. They have larger diameters as to journal, also center of axle, but their wheel seat is exactly the same dimensions as the M. C. B. 5-in. by 9-in. axle—in other words, 6½ in. Mr. Pilcher did not allow for any turning down of the wheel seat in service, where they changed the wheels under any such cars. The figures given by Mr. Pilcher are entirely correct, in accordance with the formula, that if the axle has a wheel seat of 6½ in. it is good for a load of 148,000 lb., but if the wheel seat is turned down to 6⅜ in., it is good for a load of 140,000 lb, and if turned down to 6¼ in., which is the minimum limit for the 5-in. by 9-in. axle, it is good for 132,000 lb. Axles, under the N. & W. cars referred to, have wheel seats turned down to 6⅜ in. and 6¼ in., and therefore, the committee could not consistently accept the recommendation of Mr. Pilcher.

For many years we have talked about the overloading of our refrigerator cars, considering the size of the journal that was under the car and the capacity marking on the car. Now, we all agree that the refrigerator marking was not correct considering the diameter of the axle under the car. After a while we found a lot of our cars weak and we began to apply steel underframes, steel centers and steel ends, and other improvements and added to their weight. Finally we found that our box cars had the journals overloaded with the marked capacity they were supposed to carry, considering the 10 per cent added.

Mr. Pilcher brings out the dimensions of his axle. He says it is strong. The 5½ in. by 10 in. axle was established because its dimensions were found necessary in order to carry the load that the car was supposed to carry, that it was placed under. He has not made any mention as to what the load would represent on the journal, or the journal bearing, that would increase the heating of his 5 in. by 9 in. journal over the 5½ in. by 10 in. journal. He probably didn't break the axle at the wheel seat, but probably broke the journal off because it was frequently hot.

Chairman: I would like to ask Mr. Pilcher whether any record has ever been kept of loose wheels.

Mr. Pilcher: We have loose wheels, but I am sure there are no more loose wheels in connection with this construction than any other construction we have. The loose wheel construction is not a question of the particular size of that axle fit. It is a very serious matter to cut out such a large hauling capacity in the country at this time, and the cost of replacement is so very great.

I would like, therefore, if I could get a second to my motion, so that the matter could come up for a vote, if nothing else, to see if we cannot stave off this enormous expense .

Mr. Kleine brought out the point of the turning down of the axles. Penalize us when you find them turned down, and don't penalize those that are not turned down.

There has been no indication whatever of any unexpected or any unusual heating that would bring about the breaking of these axles in the journal. We have had broken axles, due to hot boxes, but I cannot say that any one of them was this particular axle.

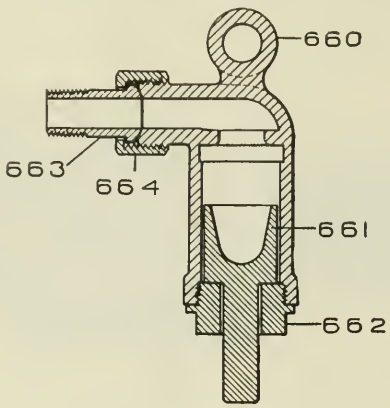
Mr. Gaines: I would like to ask the Secretary when the 5½ by 10 axle was first made the standard of the Association.

Secretary: That was in 1899, Mr. Gaines. In 1901 the diameter of the wheel seat was changed from 6⅞ to 7 in.

H. W. Coddington (N. & W.): It is my duty to make a report on all of these failures wherever they occur, and it is our experience that we have more broken journals in the boxes in

Sellers Injector Indicator

THE USUAL FORM of "tell-tale" for non-lifting injectors warns the engineer by discharging a jet of steam into the cab when the injector blows back. The indicator illustrated herewith, and which is exhibited by William Sellers & Co., Inc., not only performs the function of tell tale without discharge of steam into the cab, but, in addition, advises the engineer if the injector is wasting water at the overflow and enables the operator to adjust the feed valve to the actual maximum or minimum capacities of the injectors. Its action depends upon the well-known principle that when feeding without waste there is always a partial vacuum within the overflow chamber of an injector of the gravity overflow type. This vacuum is utilized to raise the loosely fitting piston to its upper seat, causing the projecting plug to disappear from view. If, for any cause, the injector starts to waste, the vacuum within the overflow chamber is broken, the piston drops to the lower end of its stroke and exposes the projecting plug. The operator is thus warned and will partially close the lazy cock, stopping the waste.



Injector Indicator

If there is an interruption of the water or steam supply, and the injector "flies off," the piston will instantly drop to its lower seat, preventing outflow of steam; the engineer will be again warned of the danger.

A further advantage of this form of indicator is that it enables the engineer or fireman to obtain the actual minimum capacity and adjust the feed to the varying needs of the boiler. Most operators do not regulate the water supply closely, fearing that the injector may break off without warning, or waste at the overflow. The indicator is so sensitive to conditions in the overflow chamber that the exact minimum can be obtained by regulating the lazy cock until the end of the piston appears, due to the loss of the vacuum in the overflow chamber; a very slight opening of the lazy cock will cause the piston to rise and the exact minimum is obtained, with resulting economy of fuel.

Further, the application of the indicator will reduce the tendency of the engineer to close the heater valve, which prevents the injector from restarting automatically, so that this form of indicator materially increases the safety of the boiler feeding system. The Sellers indicator makes the non-lifting injector as safe and certain to operate by night or day as an open overflow lifting injector.

It is said to be applicable to the Sellers, Nathan WF, Nathan Simplex, Chicago, and other forms of non-lifting injectors using an overflow chamber containing the combining tube and closed against the atmosphere by a gravity overflow valve.

Chrobaltic High Speed Cutters

MILLING MACHINE CUTTERS cast from high speed steel are one of the new products developed to meet war-time demands, which are being exhibited at the convention. The demand for high speed steel for munitions work taxed the available supply of tungsten, and to meet the emergency the Chrobaltic Tool Company, Chicago, developed a high speed alloy steel made without tungsten, known as Chrobaltic alloy. This new product not only makes the use of tungsten unnecessary, but also saves material and high priced labor, as it can be cast into the most intricate shapes.

Chrobaltic alloy has been used in the manufacture of milling cutters, blanking, drawing and forming dies and hot and cold trimmers. It offers remarkable resistence to abrasion and is therefore particularly adapted for gages and similar instruments. The cutter or tool is cast to practically the finished form, leaving from 1/16 in. to 1/8 in. for finish, and the tools are machined to within a narrow tolerance before heat treating. Final finish, however, is attained by grinding as in the case of regular high speed steel cutters. These new high speed steel castings have the appearance of good smooth forgings. They are practically non-rusting and acid-resisting, and their grain is uniform and dense. The heat resisting qualities of the new alloy are high and it does not scale in the fire, thus adapting it to milling cutters having very thin cutting edges. The cast tools are furnished in the finished state ready for use, or they are supplied in annealed form to the customer who machines them to accurate dimensions and hardens them for use. They may be hardened in oil or air, and the process of annealing and rehardening may be repeated many times without losing any of the properties of the steel.

Annealing is effected by heating to 1,832 deg. F. in closed muffle furnaces and allowing the part to cool slowly. However, very rapid annealing may be effected by heating slowly to 1,202 deg. F. in a muffle furnace or in a blacksmith's fire, allowing the temperature to fall there to a black heat, and afterward cooling naturally in still air or quenching in boiling water. After annealing, the casting should show about 38 scleroscope hardness. It can then be machined at the rate of 25 to 30 lineal ft. per minute.

To harden Chrobaltic cutters, they should be heated slowly in a hardening muffle furnace to 1,832 deg. F. to 1,877 deg. F. and held at this heat until the tools are thoroughly soaked. The heat is then allowed to drop about 20 to 50 deg., at which heat the cutter is withdrawn and quenched in oil. Good results may also be obtained by allowing the cutter to cool in still air until the red disappears and is invisible in the shade, then quenching in oil. The slight drop in temperature while the tool is still in the hardening furnace is important. On very large milling cutters, a blast or air current may be used to accelerate cooling before quenched in oil.

In comparing the cutting and wearing qualities of Chrobaltic alloy with those of tungsten high speed steel, there is practically no difference where the ordinary temperatures in service are involved, but in abrasive service, as in the machining of phosphor bronze, white metals, etc., as much as ten times the wear is obtainable from Chrobaltic cast cutters.

pressure for a given brake pipe reduction, as compared with the Westinghouse triple values, particularly in view of the short piston travel maintained on the Virginian Railway. In mixed trains better braking conditions would be provided and more harmonious operation with other types of brake apparatus if the A. S. A. brake provided higher brake cylinder pressure in proportion to brake pipe reduction. It is noted that this is merely a detail of construction and an increase in the present ratio can easily be obtained.

In mixed trains with the A. S. A. triple valve equipment on the head end the functions of graduated release were accomplished and tests demonstrated that the graduated release feature of the A. S. A. triple when used on the head end of mixed trains is very desirable under operating conditions similar to those encountered on the Virginian Railway, also emergency operation of the A. S. A. cars on the head end of mixed trains was obtained following a service application when the initial reduction was not too great. The use of A. S. A. triple valves in mixed trains on account of locally reducing and increasing brake pipe pressure results in some degree in more positive operation and an improvement in serial action both in applications and in releasing the brakes, the degree being dependent upon the number and location of the A. S. A. triple valves in the train.

On cars equipped with A. S. A. apparatus, the A. S. A. triple valves supplied comparatively uniform brake cylinder pressure which was not affected by piston travel, brake cylinder leakage or pressure retained in the brake cylinder at the beginning of an application; but under present practices of controlling trains the brakes ordinarily are not held applied long enough to obtain the full advantage in mixed trains of the feature of compensating for brake cylinder leakage.

In the standing tests with trains having all cars equipped with A. S. A. triple valves, practically all the intended functions were accomplished. In general, proper serial action in both application and release was obtained; when operating in graduated release the brakes were graduated on and off as desired, being fully released as intended when normal brake pipe pressure was restored. Comparatively uniform brake cylinder pressure was obtained regardless of piston travel or of pressure retained in the brake cylinders by the retaining valves, and it was maintained notwithstanding ordinary brake cylinder leakage. Emergency operation following a service application was not secured in every case when desired; after the installation of the special emergency valve on the locomotive, emergency operation was secured when the initial service reduction was 20 pounds or less. The emergency tests demonstrated that after a full service reduction followed by a full release period as short as 10 seconds emergency operation of the triples throughout the train can be secured. The automatic emergency feature of the triple valves tested, which becomes effective when the brake pipe pressure had been depleted to five pounds or less, was so much delayed as to be of little or no practical value; to make this feature of the brake system useful the period of time required before the automatic emergency operation occurs must be materially decreased, which can be accomplished by making this function operative at a higher brake pipe pressure. In the test providing for an angle cock on the rear end of the train to be opened with the brake valve in running position, the emergency operation carried to the head end of the 100-car train.

In the running tests on the grade between Princeton and Kelleysville with 50 and 100-car trains equipped exclusively with A. S. A. triple valves the operation of the brakes was precise and uniform and the control of the slack action during application and release of the brakes was practically ideal. The speed restriction for freight trains on this grade is 12 miles per hour, but was not observed in common practice. The test trains were in several cases permitted to attain speeds more than double the maximum authorized before the brakes were applied, necessitating a heavy brake application to control them. The pressure developed in the brake cylinder was comparatively uniform except on the head end, where high brake cylinder pressure resulted from an overcharge of the brake pipe while the brake valve was in full release position. The release of the brakes even at as low speed as two miles per hour resulted in no bad effects from slack action, but, on the contrary, was very uniform.

In the two cases when the train parted between Kelleysville and Rich Creek the brakes had been released on the forward portion of the train and the break-in-twos occurred between cars farther back in the train on which the brakes were still applied. In neither case was there any noticeable shock or slack action at either end of the train prior to the break-in-two. At the time of these break-in-twos all of the triple valves were operated in quick release and no retaining valves were used. None of the emergency operations on the 100-car train on the trip from Princeton to Roanoke, originating at different points in the train, caused excessive shock and no damage to train or equipment resulted, notwithstanding the low rate of speed in the first two instances. After turning the summit of the grade at Merrimac, all cars being in graduated release, an excessive brake pipe reduction was made; in attempting to release the brakes the brake valve was held in full release position long enough to fully release the brakes on the head end, but the train was stopped before release on the rear end was secured. This resulted in slack running out, and a comparatively severe shock on the rear end, but this shock was not severe enough to cause any damage. When the break-in-two occurred at Purdy the brakes had been released on the first half of the train only and the rise in brake pipe pressure had barely reached the first A. S. A. car when the train parted; this break-in-two was also clearly due to improper manipulation of the brake valve rather than to any arrangement or feature of the triple valve equipment.

The A. S. A. feed valve and the compensating valve included in the A. S. A. locomotive equipment operated as designed and intended. The tests demonstrated that it was not necessary to use any special locomotive brake equipment in order to secure proper action of the A. S. A. triple valve, but the maintaining and compensating functions of the A. S. A. triple valves can be more fully accomplished and utilized when the compensating valve on the locomotive is used. Some alteration in the detail arrangement of the compensating valve on the locomotive as used in these tests is necessary in order not to interfere with emergency operation following a service application. When the compensating valve is in service in case of the train parting or a break in train line the engineer's brake valve must be placed in emergency position in order to prevent depletion of the main reservoir pressure.

In the tests of the A. S. A. brake system particular attention was devoted to principles of design and the functions accomplished. Some improvements, particularly in regard to the feature of securing emergency operation following a service application, can be effected and the construction of the A. S. A. triple valve can be materially simplified. An application of the principle of automatic straight air brake valves to meet the conditions of passenger train breaking was not considered in connection with these tests, which were entirely with freight equipment.

The removal of the purchasing office of Morgan's Louisiana & Texas Railroad & Steamship Company from New Orleans, La., to Houston, Texas, "smacks of inexcusable favoritism" by the Railroad Administration, according to a telegram sent by Martin Behrman, mayor of New Orleans to the director general of railroads.

Report of Committee on Superheater Locomotives

THE COMMITTEE SUBMITS the following report on superheated locomotives:

Superheating Existing Locomotives

The superheat schedule was somewhat delayed during the past few years, due to the shortage of labor and material caused by the war. Of the railroads reporting, practically all are going to the piston valve when superheating, either by changing to piston valve cylinders or applying a piston valve steam chest. One railroad, in superheating cross-compound locomotives, retained the slide valve on the low pressure side, using a double ported slide valve. Those reporting as retaining slide valves used bronze valves. In applying new cylinders new modern valve gears were applied to the locomotives. Considerable trouble was experienced in using the piston valve steam chest due to leaks and rapid wear. These defects have been overcome in the later designs. When superheat valve stem packing is used the life of the packing is about the same as that of a saturated locomotive.

Lubrication of Superheated Locomotives

On the railroads using slide valves no change was made in the oiling or lubricating devices when bronze valves were used, but where the original cast-iron valves were used some trouble with cutting the seats was experienced. The application of an additional system of graphite lubrication stopped the trouble.

The hydrostatic lubricator is still the standard for all locomotives and results from its use are satisfactory in most cases. A number of the railroads have taken up force feed lubrication, using a plunger type of lubricator. As this practice of lubrication is very recent and data as to the performance limited, the committee does not feel justified in making any recommendation, but it is the opinion of the railroads using them that the force feed lubricator when in good condition will distribute the oil more economically, positively and regularly than the hydrostatic lubricator. It has the advantage of being outside the cab and requiring no special attention of the engineman, as the feeds are set and the lubricator stops feeding when the locomotive stops. Sufficient information is not at hand to judge as to the life of packing rings and bushings; it is the opinion of the committee that the life will be increased, but is unable to say to what extent.

Cylinder feeds are not extensively used and the committee is of the opinion that they are unnecessary and should be discontinued, as it is doubtful if any benefits are derived from their use.

The use of superheat valve oil is recommended by the majority of railroads reporting, but a large prcentage use the ordinary valve oil and claim satisfactory results. Carbonization is caused by the manner of operation.

Considerable trouble has been experienced with cylinder and valve packing. Much of this trouble has been overcome by changing the design of the rings, which, in most cases, consisted of going from a ring, with a square cross-section, to a narrow-faced ring, by using a better grade of material, and by admitting steam to the cylinders while drifting. It is the opinion of the committee that two cylinder packing rings are ample for good service.

The majority of railroads report satisfactory results from a 50 per cent lead, 50 per cent copper mixture for piston rod packing. Very good results have been obtained from 74 per cent lead, 20 per cent copper and 6 per cent nickel, and 80 per cent lead and 20 per cent antimony, especially when equipped with a hard grease piston rod lubricator, which adds greatly to the life of the softer packing. Some railroads are using tandem piston rod packing, but it is the opinion of the committee that single packing will give satisfactory results.

When automatic or manually operated drifting valves, or drifting throttles are not used, instructions have been issued to enginemen to drift with what is called a cracked throttle. It is essential that steam be supplied to the cylinders while drifting. Drifting valves are not essential, but desirable, especially in a mountainous country, as the steam can be supplied by cracking the throttle. Automatic drifting valves of a good design are valuable as they eliminate the human element in furnishing steam for drifting.

A number of roads are using a manually operated drifting throtte, others are using and experimenting with automatic valves of different makes. Fifty per cent of the roads reporting do not use steam chest relief valves, and 50 per cent do when engines are not equipped with drifting valves or vacuum breakers. It is the opinion of the committee that relief valves should be used on large power if not equipped with drifting valves or vacuum breakers. It is the opinion of the committee that a properly designed drifting valve will eliminate carbon deposit, aid in lubrication, and increase the life of packing.

Maintenance

The superheater units have failed mostly at the rear return bend, the welded type giving the most trouble. The application of cast steel return bends has reduced this trouble. Some failures have occurred at the front end under the ball joint. This failure is due mainly to cinder wear, as the unit lies in the path of the cinders drawn through the tubes at a high velocity. To protest against this wear, a shield of thin steel is spot welded to the unit. Failures at this point are repaired by cutting off the worn part and welding on a new piece. No particular difference has been noted in the cinder wear of a stoker fired and hand fired locomotive. The abrasive effect of the cinders varies greatly with different coals.

The tools used to repair and maintain the units are those recommended by the Locomotive Superheater Company. When units are removed, the joints should be reground and individually tested before being replaced, and collectively after being replaced. A periodical test might be desirable, but the committee feels that it is not essential.

The majority of railroads purchase the header bolts. One uses .988 per cent carbon steel heated to between 1,400 and 1,500 degrees F. and plunged into an oil bath, then reheated and allowed to cool naturally. Another specifies that the material shall have a tensile strength of 90,000 lb. per sq. in., elastic limit not less than 65,000 per sq. in., and an elongation in 2 in. not less than 18 per cent. The metal should have the following chemical properties: carbon, .45 to .60 per cent, manganese, not over .70 per cent, sulphur not over .05 per cent, phosphorus not over .05 per cent. Some use a high grade iron without any heat treatment.

The investigation shows that in most cases a limit percentage of weight for the scrapping of superheater flues has not been established. Some give the limit percentage as from 10 per cent to 25 per cent reduction. Some give flues the hammer test. Superheater flues are reclaimed by welding up pits and holes worn in by steam leaks. It is a good practice to weld the large superheater flues to the back flue sheet. Good results have been obtained from it, and the committee recommends this practice.

All boiler tubes should be blown out each time the fire is drawn and, at least, at each washout period all tubes should be cleaned thoroughly from end to end, and all accumulations removed; special attention should be given to the superheater flues. The flues should be blown with a ¼ in. or ⅜ in. pipe and 100 lb. of air.

The method employed in testing the packing for leaks around the outside steam pipes where they enter the smoke box is by applying a lighted torch to all surface joints while the blower is on. Much trouble is experienced in keeping these joints tight. Different kinds of packings have been used with varied results. Cement and ground magnesia have been used with some success. A mixture of ground magnesia and asphaltum applied while hot gives promise of better results; this is due to the heat keeping the mixture in a semi-plastic state. The design as used on the United States standard locomotives gives promise of a high efficiency.

Most railroads have issued through bulletins or personally by the road foreman, instructions not to carry over two gages of water; this, of course, depends on the water conditions of the locality in which the locomotive is operating. The use of a pyrometer will show the men the low temperature of the steam resulting from carrying high water; it will also show the effect the position of the throttle and reverse lever has on the superheat obtained. The majority of the roads prefer a wide open